

MAJOR PROJECT REPORT
ON
“Optimization of Bank Telemarketing Campaigns
through Analytics”

Submitted by
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Under the supervision of
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IPU

GURU GOBIND SINGH
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CERTIFICATE

This is to certify that Ms. Tanya Mathpal registered in the MBA(Analytics) programme of USMS, GGSIPU is permitted to carry out the proposed work presented in this Major Project Report under my guidance. The matter embodied in this proposal has not been submitted earlier.

Date:

Signature

(Dr. Praveen Kumar)

DECLARATION

I hereby declare that matter embodied in this Summer Internship Report entitled **“Optimization of Bank Telemarketing Campaigns through Data-Driven Analytics”** submitted to USMS, GGSIPU is a record of original work done by me, under the guidance and supervision of Dr. Praveen Kumar and it has not formed the basis for the award of any Degree/Diploma/Associateship/ Fellowship or other similar title to any candidate of any University.

Tanya Mathpal

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Lastly, I wish to express my heartfelt thanks to my family and friends for their constant encouragement, understanding, and moral support throughout this Project.

EXECUTIVE SUMMARY

In order to improve the effectiveness of retail banking outreach, the project report "Optimization of Bank Telemarketing Campaigns through Data-Driven Analytics" investigates the strategic integration of data science and marketing. The study uses a large dataset of more than 11,000 customer records to solve the fundamental problems of traditional, volume-based telemarketing, such as high operating costs and low conversion rates. The study finds important demographic and behavioral indicators, such as account balance, job status, and call time, that have a major impact on a customer's inclination to sign up for a term deposit by using rigorous exploratory data analysis and performance attribution.

A simulated A/B test comparing various communication channels is a key part of the study, and it unequivocally demonstrates that cellular contact techniques produce better conversion rates than conventional landlines. Additionally, the analysis highlights how data-driven lead scoring may significantly lower cost-per-acquisition while optimizing return on investment by quantifying the relationship between engagement quality and successful results.

The results indicate that financial institutions can improve client connections and generate more financial growth by switching from broad-spectrum marketing to a tailored, analytics-based methodology. This paper concludes that sustaining a competitive edge in the changing financial services sector requires a synergy between predictive analytics and strategic execution, and it ultimately serves as a blueprint for modernizing bank marketing operations. In addition to increasing the bank's profitability, this change guarantees a more customized and unobtrusive customer experience.

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CHAPTER 1

INTRODUCTION

1.1 Overview of the Banking Sector and Marketing

The retail banking sector has transitioned from a conventional service-based approach to a fiercely competitive, data-driven environment. In this context, the capacity to successfully cross-sell products, like term deposits, is essential for sustaining liquidity and nurturing lasting customer relationships. Nonetheless, banks consistently grapple with the issue of "marketing fatigue," where customers become inundated with frequent and often irrelevant messages. To address this, financial institutions are increasingly adopting data-focused strategies to ensure the appropriate message is delivered to the right customer at the ideal moment.

1.2 The Digital Age of Telemarketing

One of the most straightforward and intimate methods for bank outreach is still telemarketing. A phone-based campaign, in contrast to passive advertising, enables instantaneous feedback and real-time involvement. Telemarketing requires a lot of resources, including substantial expenditures for operational infrastructure and human capital, notwithstanding its efficacy. The accuracy with which potential subscribers are identified is just as important to the success of these campaigns as the quantity of calls made.

1.3 Problem Description

The ineffectiveness of "blind" telemarketing efforts is the main issue this research attempts to solve. High operating expenses, low conversion rates, and possibly unhappy clients who feel hounded by irrelevant offers are the outcomes of a bank targeting its whole customer base without first doing an analysis. Analyzing past campaign data is crucial to determining the behavioral and demographic indicators that are associated with a "Yes" response (term deposit subscription).

The bank can switch from a volume-based strategy to a value-based strategy by recognizing these trends, concentrating efforts on high-probability prospects.

1.4 The Dataset

The dataset contains a variety of data, such as:

- Customer demographics include age, education, marital status, and type of employment.
- Financial indicators include current personal or housing loans, average annual debt, and default history.
- Contact type (cellular or telephone), day, month, and length of the most recent contact are examples of communication metadata.
- Campaign History: The quantity of contacts made during this campaign and the results of earlier marketing initiatives.

OBJECTIVE

- Evaluating conversion rates broken down by customer profile to prioritize segments
- Determining reach out factors that works best (timing, channel, duration)
- A predictive model that scores customers by conversion likelihood
- To Evaluate Communication Channel Effectiveness
- ROI analysis with respect to cost per accusation, which customer segments are worth the cost of contact

CHAPTER 2

LITERATURE REVIEW

In the banking industry, traditional marketing techniques frequently depended on intuition-based decision-making and broad audience targeting. However, because of their significant resource misallocation, such static structures are increasingly seen as unsustainable and wasteful (Eman, 2025). Real-time data analytics has become more prevalent in today's financial environment, allowing for quick, evidence-based changes. According to research, using these tools can significantly increase user involvement and strategic decision-making (Eman, 2025).

The capacity to forecast consumer behaviour prior to the commencement of a campaign is the foundation of telemarketing optimization. According to Vashi et al. (2024), it is no longer practical for human analysts to handle the massive amounts of electronic data produced by clients. Instead, whether a consumer will sign up for a term deposit is predicted using data mining algorithms on customer demographics, such as age, employment, and loan status (Vashi et al., 2024).

To improve prediction reliability, ensemble learning techniques such bagging, boosting, and stacking have been introduced recently (Peter et al., 2025). Research shows that stacking models can distinguish between prospective subscribers and non-subscribers with accuracy rates as high as 91.88% (Peter et al., 2025). By efficiently handling heterogeneous data, other frameworks, including the Class Membership-Based (CMB) classifier, have demonstrated even greater performance marks, reaching up to 97.3% accuracy (Tjouabou et al., 2022).

The factors that most affect a telemarketing call's outcome are regularly identified in the literature:

- **Contact Duration:** According to several studies (Peter et al., 2025; Tjouabou et al., 2022), this is the most important element in predicting the chance of a subscription.
- **Economic Indicators:** The actual economic climate in which telemarketing operates is reflected in variables like the Euribor rate and macroeconomic swings (Peter et al., 2025).
- **Customer Demographics:** Several predictive frameworks continue to include age and job type as fundamental indicators (Vashi et al., 2024).

"Class imbalance," where the bulk of contacts (often over 88%) end in a non-subscription, is a key issue seen in telemarketing datasets (MDPI, 2026). Standard machine learning models have a tendency to over predict the majority class in the absence of appropriate modification, which results in poor recall of real subscribers. To solve these fundamental issues and guarantee that models continue to be reliable, researchers suggest sophisticated balancing strategies including cost-sensitive learning and the Synthetic Minority Over-Sampling Technique (SMOTE) (MDPI, 2026; Peter et al., 2025).

Banks utilize A/B testing (or split testing) to establish causation, or if a certain campaign tweak actually led to a conversion. In order to isolate the impact of a single modification, banks can use this experimental strategy to test two versions (such as cellular vs. telephone contact) under similar settings (Logos Science, 2026). Although A/B testing is the "gold standard" for proving causation, current research indicates that it works best when combined with machine learning to forecast performance outcomes using display features from the past (Research@CBS, n.d.).

Enhancing the Return on Investment (ROI) is the ultimate objective of data-driven optimization. In order to ensure that marketing expenditures are focused on high-quality leads, predictive analytics allows banks to proactively handle possible problems, such as identifying clients who are likely to churn or who have high credit risk (Logos Science, 2026; Multi research Journal, 2025).

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Design

A quantitative and analytical research design is used in this study. It uses a diagnostic method to assess the efficacy of different marketing levers and a descriptive and exploratory approach to comprehend consumer behavior. The approach is designed to use statistical analysis and data visualization to move from the input of raw data to meaningful business knowledge.

3.2 Data Collection

The "Bank Marketing Dataset," which includes details on a financial institution's direct marketing activities (phone calls), serves as the project's main data source.

- 11,162 observations make up the data size.
- Features: 17 variables that include campaign-specific metadata (length, contact type, month), financial data (amount, loan status), and demographic data (age, employment, marital status).
- 'deposit' is the target variable, which indicates if the client has a term deposit subscription.

3.3 Preparing and Cleaning Data

To guarantee accuracy and dependability, the data underwent a thorough cleaning procedure prior to analysis:

- Missing Value Analysis: To avoid biased findings, look for null entries in each of the 17 columns.
- Data Transformation: The frequency and distribution of categorical variables like "job," "education," and "contact" were evaluated.
- Finding extreme values in numerical fields like "balance" and "duration" that might distort the statistical mean is known as outlier detection.

3.4 Methods and Instruments for Data Analysis

Python-based data science libraries were used in a Jupyter Notebook environment to do the analysis. Among the essential tools were:

- Pandas and NumPy: For structural analysis and data processing.
- For creating descriptive visualizations (box plots, and histograms), use Matplotlib with Seaborn.
- Statistical Libraries: For simulating A/B tests and computing conversion rates.

3.5 Framework for Analysis

The following four-stage structure was used to carry out the research:

- Analysis of Exploratory Data (EDA)

A thorough analysis of the data to find correlations. This required examining the correlation between consumer demographics (such as age or kind of job) and subscription likelihood.

- Attribution of Performance

This step entailed figuring out which particular marketing elements result in the highest conversion. We examined factors like "month of contact" and "number of previous contacts" to determine how they affected the outcome.

- A/B Testing Simulation

The modeling of an A/B test to assess contact tactics was a fundamental part of the process. Customers contacted by conventional landline (telephone) comprise the Control Group(Group A). Customers reached by mobile (cellular) are the test group (group B). The best communication channel was identified by comparing the conversion rates of the two groups.

- Cost and ROI Analysis

A framework for determining the "Cost Per Acquisition" (CPA) and "Return on Investment" (ROI) by allocating simulated expenses to call durations and operational overheads was part of the methodology.

CHAPTER 4

DATA ANALYSIS

```
import sqlite3, warnings, os
import numpy          as np
import pandas         as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn        as sns
import plotly.express  as px
import plotly.graph_objects as go
from plotly.subplots  import make_subplots
import plotly.io       as pio
from scipy            import stats
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics    import roc_auc_score, classification_report
```

Update path based on path finder output above

```
df = pd.read_csv("bank.csv")
print(f'Loaded: {df.shape[0]:,} rows x {df.shape[1]} columns')
print()
print(df.dtypes.to_string())
print()
print('Conversion rate:')
print(df['deposit'].value_counts(normalize=True).mul(100).round(1).to_string())
```

Loaded: 11,162 rows x 17 columns

age	int64
job	object
marital	object
education	object
default	object
balance	int64
housing	object
loan	object
contact	object
day	int64
month	object
duration	int64
campaign	int64
pdays	int64
previous	int64
poutcome	object
deposit	object

Conversion rate:

deposit	
no	52.6
yes	47.4

```
print('=== NULL AUDIT ===')
nulls = pd.DataFrame({
    'dtype':      df.dtypes,
    'null_count': df.isnull().sum(),
    'unique':     df.nunique(),
    'sample':     df.iloc[0],
})
print(nulls.to_string())
```

=== NULL AUDIT ===

	dtype	null_count	unique	sample
age	int64	0	76	59
job	object	0	12	admin.
marital	object	0	3	married
education	object	0	4	secondary
default	object	0	2	no
balance	int64	0	3805	2343
housing	object	0	2	yes
loan	object	0	2	no
contact	object	0	3	unknown
day	int64	0	31	5
month	object	0	12	may
duration	int64	0	1428	1042
campaign	int64	0	36	1
pdays	int64	0	472	-1
previous	int64	0	34	0
poutcome	object	0	4	unknown
deposit	object	0	2	yes

```
df['deposit_binary'] = (df['deposit'] == 'yes').astype(int)

conn = sqlite3.connect(':memory:')
df.to_sql('campaigns', conn, index=False, if_exists='replace')
print('Loaded into SQLite.')
print(f'Total contacts: {len(df):,}')
print(f'Overall conversion rate: {df["deposit_binary"].mean()*100:.1f}%')
```

Loaded into SQLite.
Total contacts: 11,162
Overall conversion rate: 47.4%

```

print('=== CONVERSION RATE BY JOB TYPE ===')
sql1 = (
    'SELECT job,'
    ' COUNT(*) as contacts,'
    ' SUM(deposit_binary) as conversions,'
    ' ROUND(CAST(SUM(deposit_binary) AS FLOAT)/COUNT(*)*100, 1) as conversion_rate'
    ' FROM campaigns GROUP BY job ORDER BY conversion_rate DESC'
)
print(pd.read_sql_query(sql1, conn).to_string(index=False))

print()
print('=== CONVERSION RATE BY EDUCATION ===')
sql2 = (
    'SELECT education,'
    ' COUNT(*) as contacts,'
    ' SUM(deposit_binary) as conversions,'
    ' ROUND(CAST(SUM(deposit_binary) AS FLOAT)/COUNT(*)*100, 1) as conversion_rate'
    ' FROM campaigns GROUP BY education ORDER BY conversion_rate DESC'
)
print(pd.read_sql_query(sql2, conn).to_string(index=False))

```

=== CONVERSION RATE BY JOB TYPE ===

job	contacts	conversions	conversion_rate
student	360	269	74.7
retired	778	516	66.3
unemployed	357	202	56.6
management	2566	1301	50.7
unknown	70	34	48.6
admin.	1334	631	47.3
self-employed	405	187	46.2
technician	1823	840	46.1
services	923	369	40.0
housemaid	274	109	39.8
entrepreneur	328	123	37.5
blue-collar	1944	708	36.4

=== CONVERSION RATE BY EDUCATION ===

education	contacts	conversions	conversion_rate
tertiary	3689	1996	54.1
unknown	497	252	50.7
secondary	5476	2450	44.7
primary	1500	591	39.4

```

print('=== CONVERSION RATE BY CONTACT MONTH ===')
sql3 = (
    'SELECT month,'
    ' COUNT(*) as contacts,'
    ' SUM(deposit_binary) as conversions,'
    ' ROUND(CAST(SUM(deposit_binary) AS FLOAT)/COUNT(*)*100, 1) as conversion_rate'
    ' FROM campaigns GROUP BY month ORDER BY conversion_rate DESC'
)
print(pd.read_sql_query(sql3, conn).to_string(index=False))

print()
print('=== CONVERSION BY NUMBER OF PREVIOUS CONTACTS ===')
sql4 = (
    'SELECT'
    ' CASE'
    '   WHEN campaign = 1 THEN "1 contact"'
    '   WHEN campaign = 2 THEN "2 contacts"'
    '   WHEN campaign <= 5 THEN "3 to 5 contacts"'
    '   ELSE "6+ contacts"'
    ' END as contact_frequency,'
    ' COUNT(*) as customers,'
    ' ROUND(CAST(SUM(deposit_binary) AS FLOAT)/COUNT(*)*100, 1) as conversion_rate'
    ' FROM campaigns GROUP BY contact_frequency ORDER BY conversion_rate DESC'
)
print(pd.read_sql_query(sql4, conn).to_string(index=False))

```

=== CONVERSION RATE BY CONTACT MONTH ===

month	contacts	conversions	conversion_rate
dec	110	100	90.9
mar	276	248	89.9
sep	319	269	84.3
oct	392	323	82.4
apr	923	577	62.5
feb	776	441	56.8
aug	1519	688	45.3
jun	1222	546	44.7
nov	943	403	42.7
jul	1514	627	41.4
jan	344	142	41.3
may	2824	925	32.8

=== CONVERSION BY NUMBER OF PREVIOUS CONTACTS ===

contact_frequency	customers	conversion_rate
1 contact	4798	53.4
2 contacts	3028	46.3
3 to 5 contacts	2470	43.5
6+ contacts	866	29.2

```

print('=== CALL DURATION vs CONVERSION (SQL BUCKETS) ===')
sql5 = (
    'SELECT'
    ' CASE'
    '   WHEN duration < 60   THEN "Under 1 min"'
    '   WHEN duration < 180 THEN "1 to 3 mins"'
    '   WHEN duration < 300 THEN "3 to 5 mins"'
    '   WHEN duration < 600 THEN "5 to 10 mins"'
    '   ELSE "Over 10 mins"'
    ' END as call_duration_bucket,'
    ' COUNT(*) as contacts,'
    ' ROUND(CAST(SUM(deposit_binary) AS FLOAT)/COUNT(*)*100, 1) as conversion_rate,'
    ' ROUND(AVG(duration), 0) as avg_duration_secs'
    ' FROM campaigns'
    ' GROUP BY call_duration_bucket'
    ' ORDER BY avg_duration_secs'
)
print(pd.read_sql_query(sql5, conn).to_string(index=False))

```

```

=== CALL DURATION vs CONVERSION (SQL BUCKETS) ===
call_duration_bucket  contacts  conversion_rate  avg_duration_secs
           Under 1 min         715             1.3             34.0
           1 to 3 mins        3203            21.6            121.0
           3 to 5 mins        2436            46.0            235.0
           5 to 10 mins        2650            61.5            422.0
           Over 10 mins        2158            85.1            949.0

```

```

# Conversion rate by key categorical features
cat_features = ['job', 'marital', 'education', 'contact', 'month', 'outcome']

fig, axes = plt.subplots(2, 3, figsize=(18, 11))
fig.patch.set_facecolor('white')
axes = axes.flatten()

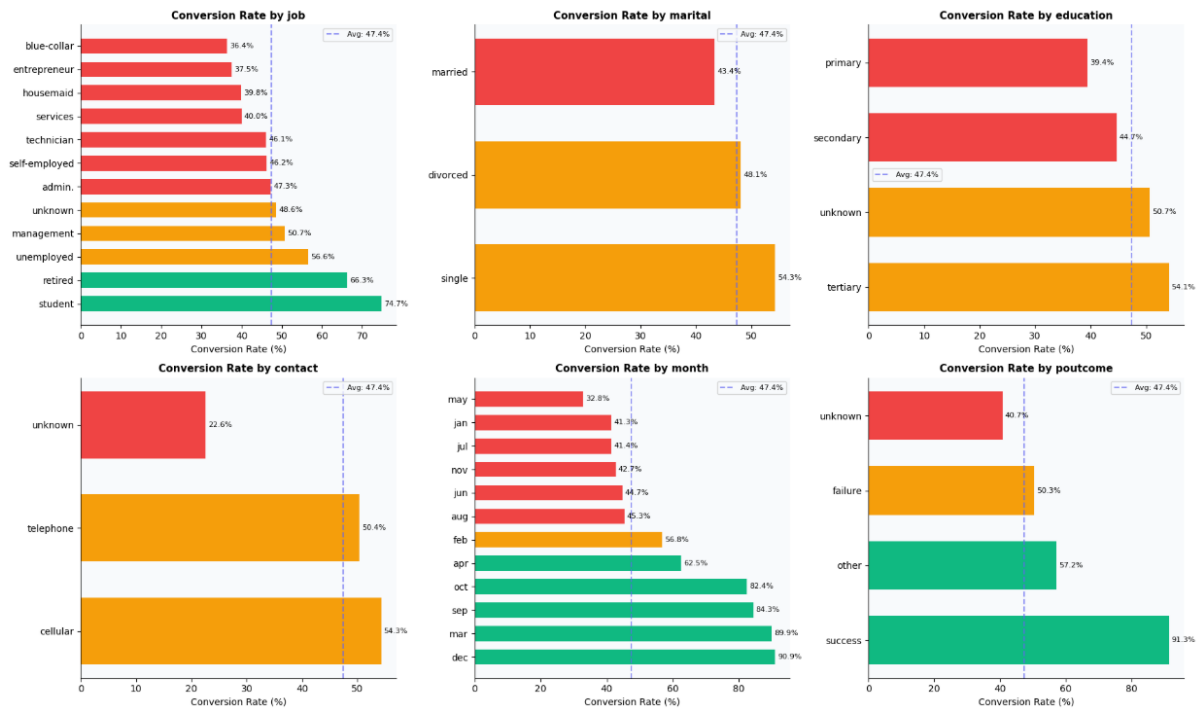
overall_rate = df['deposit_binary'].mean() * 100

for ax, col in zip(axes, cat_features):
    rates = (
        df.groupby(col)['deposit_binary']
        .mean().mul(100).round(1)
        .sort_values(ascending=False)
        .reset_index()
    )
    bar_colors = [
        C['yes'] if v >= overall_rate * 1.2
        else C['amber'] if v >= overall_rate
        else C['no']
        for v in rates['deposit_binary']
    ]
    bars = ax.barh(
        rates[col].astype(str),
        rates['deposit_binary'],
        color=bar_colors, edgcolor='none', height=0.65
    )
    ax.axvline(overall_rate, color=C['primary'], linestyle='--',
               linewidth=1.5, alpha=0.7, label=f'Avg: {overall_rate:.1f}%')
    ax.set_title(f'Conversion Rate by {col}', fontweight='bold', fontsize=11)
    ax.set_xlabel('Conversion Rate (%)')
    ax.set_facecolor(C['bg'])
    ax.bar_label(bars, fmt='%.1f%%', padding=3, fontsize=8)
    ax.legend(fontsize=8)

plt.suptitle('Campaign Conversion Rates by Customer and Contact Attributes',
             fontsize=14, fontweight='bold', y=1.01)
plt.tight_layout()
plt.savefig('/kaggle/working/output/conversion_by_feature.png', dpi=120, bbox_inches='tight')
plt.show()

```


Campaign Conversion Rates by Customer and Contact Attributes



Call duration vs conversion - the key relationship

```
fig, axes = plt.subplots(1, 2, figsize=(16, 5))
```

```
fig.patch.set_facecolor('white')
```

Box plot: call duration by outcome

```
converted = df[df['deposit'] == 'yes']['duration'] / 60
```

```
not_converted = df[df['deposit'] == 'no']['duration'] / 60
```

```
bp = axes[0].boxplot(
    [not_converted.clip(upper=not_converted.quantile(0.95)),
     converted.clip(upper=converted.quantile(0.95))],
    labels=['Not Converted', 'Converted'],
    patch_artist=True,
    medianprops=dict(color=C['primary'], linewidth=2),
    whiskerprops=dict(color=C['subtext']),
    capprops=dict(color=C['subtext']),)
```

Safely colour the boxes by index

```
box_colors = [C['no'] + '55', C['yes'] + '55']
```

```
for patch, color in zip(bp['boxes'], box_colors):
    patch.set_facecolor(color)
```

```
axes[0].set_title('Call Duration by Conversion Outcome', fontweight='bold')
```

```
axes[0].set_ylabel('Call Duration (minutes)')
```

```
axes[0].set_facecolor(C['bg'])
```

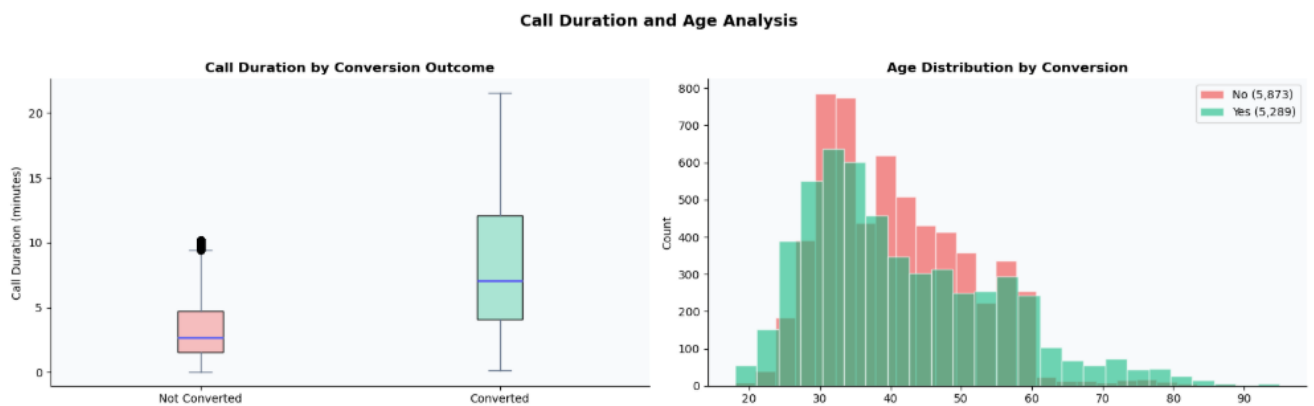
```

# Age distribution by conversion
for label, color in [('no', C['no']), ('yes', C['yes'])]:
    subset = df[df['deposit'] == label]['age']
    axes[1].hist(subset, bins=25, alpha=0.6, color=color,
                  label=f'{label.title()} ({len(subset):,})', edgecolor='white')
axes[1].set_title('Age Distribution by Conversion', fontweight='bold')
axes[1].set_xlabel('Age')
axes[1].set_ylabel('Count')
axes[1].legend(fontsize=10)
axes[1].set_facecolor(C['bg'])

plt.suptitle('Call Duration and Age Analysis', fontsize=14, fontweight='bold', y=1.02)
plt.tight_layout()
plt.savefig('/kaggle/working/output/duration_age.png', dpi=120, bbox_inches='tight')
plt.show()

t_stat, p_val = stats.ttest_ind(converted, not_converted)
print(f'T-test: call duration difference is statistically significant')
print(f'Converted avg: {converted.mean():.1f} minutes')
print(f'Not converted avg: {not_converted.mean():.1f} minutes')
print(f'p-value: {p_val:.6f} (p < 0.05 = significant)')

```



T-test: call duration difference is statistically significant
 Converted avg: 9.0 minutes
 Not converted avg: 3.7 minutes
 p-value: 0.000000 (p < 0.05 = significant)

Simulated A/B Test

- **Group A (Control):** Customers contacted by telephone
- **Group B (Treatment):** Customers contacted by cellular

Null hypothesis: There is no difference in conversion rate between telephone and cellular contact.

Alternative hypothesis: Cellular contact produces a significantly different conversion rate.

```
# Filter to telephone vs cellular (the natural A/B split in this dataset)
group_a = df[df['contact'] == 'telephone']['deposit_binary']
group_b = df[df['contact'] == 'cellular']['deposit_binary']

print('=== A/B TEST SETUP ===')
print(f'Group A (Telephone): {len(group_a):,} contacts')
print(f'Group B (Cellular): {len(group_b):,} contacts')
print()
print(f'Group A conversion rate: {group_a.mean()*100:.2f}%')
print(f'Group B conversion rate: {group_b.mean()*100:.2f}%')
print(f'Absolute difference: {(group_b.mean()-group_a.mean())*100:.2f} percentage points')
print(f'Relative lift: {((group_b.mean()-group_a.mean())/group_a.mean()*100):.1f}%')
print()
# Chi-square test
contingency = pd.crosstab(df['contact'], df['deposit'])
chi2, p_value, dof, expected = stats.chi2_contingency(contingency)

print('=== CHI-SQUARE TEST RESULTS ===')
print(f'Chi-square statistic: {chi2:.4f}')
print(f'Degrees of freedom: {dof}')
print(f'P-value: {p_value:.6f}')
print()
if p_value < 0.05:
    print('RESULT: Statistically significant (p < 0.05)')
    print('We REJECT the null hypothesis.')
    print('The difference in conversion rate is NOT due to chance.')
else:
    print('RESULT: Not statistically significant (p >= 0.05)')
    print('We FAIL to reject the null hypothesis.')
    print('The difference may be due to random variation.')
```

=== A/B TEST SETUP ===

Group A (Telephone): 774 contacts

Group B (Cellular): 8,042 contacts

Group A conversion rate: 50.39%

Group B conversion rate: 54.33%

Absolute difference: 3.94 percentage points

Relative lift: 7.8%

=== CHI-SQUARE TEST RESULTS ===

Chi-square statistic: 736.6867

Degrees of freedom: 2

P-value: 0.000000

RESULT: Statistically significant ($p < 0.05$)

We REJECT the null hypothesis.

The difference in conversion rate is NOT due to chance.

```
# A/B test visualisation
from statsmodels.stats.proportion import proportion_confint

ci_a = proportion_confint(group_a.sum(), len(group_a), alpha=0.05)
ci_b = proportion_confint(group_b.sum(), len(group_b), alpha=0.05)

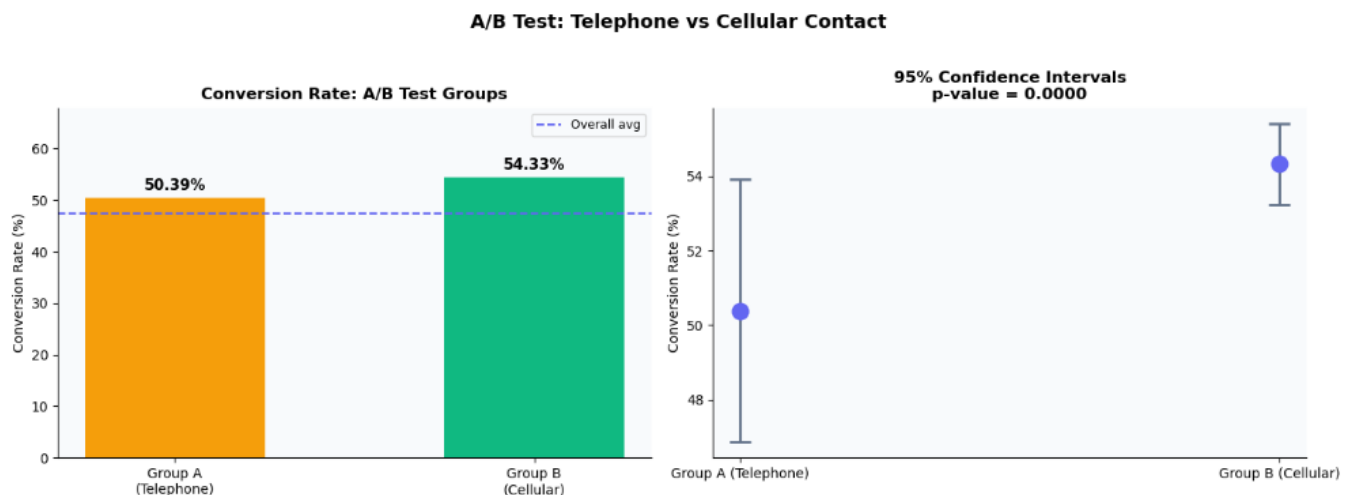
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.patch.set_facecolor('white')

groups = ['Group A\n(Telephone)', 'Group B\n(Cellular)']
rates = [group_a.mean()*100, group_b.mean()*100]
bar_cols = [C['amber'], C['yes'] if group_b.mean() > group_a.mean() else C['no']]

bars = axes[0].bar(groups, rates, color=bar_cols, edgecolor='none', width=0.5)
axes[0].axhline(df['deposit_binary'].mean()*100, color=C['primary'],
               linestyle='--', linewidth=1.5, label='Overall avg')
axes[0].set_title('Conversion Rate: A/B Test Groups', fontweight='bold')
axes[0].set_ylabel('Conversion Rate (%)')
axes[0].bar_label(bars, fmt='%0.2f%%', padding=4, fontsize=11, fontweight='bold')
axes[0].legend(fontsize=9)
axes[0].set_facecolor(C['bg'])
axes[0].set_ylim(0, max(rates) * 1.25)

axes[1].errorbar(
    ['Group A (Telephone)', 'Group B (Cellular)'],
    [group_a.mean()*100, group_b.mean()*100],
    yerr=[
        [(group_a.mean()-ci_a[0])*100, (group_b.mean()-ci_b[0])*100],
        [(ci_a[1]-group_a.mean())*100, (ci_b[1]-group_b.mean())*100]
    ],
    fmt='o', markersize=12, capsize=8, capthick=2, linewidth=2,
    color=C['primary'], ecolor=C['subtext'],)
axes[1].set_title(f'95% Confidence Intervals\np-value = {p_value:.4f}',
                  fontweight='bold')
axes[1].set_ylabel('Conversion Rate (%)')
axes[1].set_facecolor(C['bg'])

plt.suptitle('A/B Test: Telephone vs Cellular Contact', fontsize=14, fontweight='bold', y=1.02)
plt.tight_layout()
plt.savefig('/kaggle/working/output/ab_test.png', dpi=120, bbox_inches='tight')
plt.show()
```



Conversion Probability Model

```
df_model = df.copy()

# Encode categoricals
le = LabelEncoder()
cat_cols = df_model.select_dtypes(include='object').columns.tolist()
cat_cols = [c for c in cat_cols if c != 'deposit']

for col in cat_cols:
    df_model[col] = le.fit_transform(df_model[col].astype(str))

X = df_model.drop(columns=['deposit', 'deposit_binary'])
y = df_model['deposit_binary']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s = scaler.transform(X_test)

model = LogisticRegression(max_iter=1000, random_state=42, class_weight='balanced')
model.fit(X_train_s, y_train)

y_prob = model.predict_proba(X_test_s)[:, 1]
y_pred = model.predict(X_test_s)
auc = roc_auc_score(y_test, y_prob)

print(f'Model ROC-AUC: {auc:.4f}')
print()
print(classification_report(y_test, y_pred, target_names=['No', 'Yes']))
```

Model ROC-AUC: 0.8731

	precision	recall	f1-score	support
No	0.81	0.80	0.81	1175
Yes	0.78	0.79	0.79	1058
accuracy			0.80	2233
macro avg	0.80	0.80	0.80	2233
weighted avg	0.80	0.80	0.80	2233

ROI and Cost Per Acquisition by Segment

```
COST_PER_CONTACT = 5 # GBP per call
REVENUE_PER_CONV = 150 # estimated average product revenue per conversion

# Score all customers
X_all = df_model.drop(columns=['deposit', 'deposit_binary'])
X_all_s = scaler.transform(X_all)
df['conv_probability'] = model.predict_proba(X_all_s)[: , 1]

# CPA analysis by job segment
cpa_by_job = (
    df.groupby('job')
    .agg(
        contacts      = ('deposit_binary', 'count'),
        conversions    = ('deposit_binary', 'sum'),
        conversion_rate = ('deposit_binary', 'mean'),
        avg_conv_prob  = ('conv_probability', 'mean'),
    )
    .reset_index()
)
cpa_by_job['total_cost'] = cpa_by_job['contacts'] * COST_PER_CONTACT
cpa_by_job['total_revenue'] = cpa_by_job['conversions'] * REVENUE_PER_CONV
cpa_by_job['cpa'] = (cpa_by_job['total_cost'] / cpa_by_job['conversions']).round(2)
cpa_by_job['roi_pct'] = (
    (cpa_by_job['total_revenue'] - cpa_by_job['total_cost'])
    / cpa_by_job['total_cost'] * 100
).round(1)
cpa_by_job['conversion_rate'] = (cpa_by_job['conversion_rate'] * 100).round(1)
cpa_by_job = cpa_by_job.sort_values('roi_pct', ascending=False)

print('CPA and ROI by Job Segment:')
print(cpa_by_job[['job', 'contacts', 'conversions', 'conversion_rate', 'cpa', 'roi_pct']].to_string(index=False))
```

CPA and ROI by Job Segment:

job	contacts	conversions	conversion_rate	cpa	roi_pct
student	360	269	74.7	6.69	2141.7
retired	778	516	66.3	7.54	1889.7
unemployed	357	202	56.6	8.84	1597.5
management	2566	1301	50.7	9.86	1421.0
unknown	70	34	48.6	10.29	1357.1
admin.	1334	631	47.3	10.57	1319.0
self-employed	405	187	46.2	10.83	1285.2
technician	1823	840	46.1	10.85	1282.3
services	923	369	40.0	12.51	1099.3
housemaid	274	109	39.8	12.57	1093.4
entrepreneur	328	123	37.5	13.33	1025.0
blue-collar	1944	708	36.4	13.73	992.6

```

fig = make_subplots(rows=1, cols=2,
    subplot_titles=(
        'Cost Per Acquisition by Job Segment',
        'ROI (%) by Job Segment',),
    specs=[[{'type':'xy'},{'type':'xy'}]],
    horizontal_spacing=0.14,)

cpa_colors = [C['yes'] if v <= cpa_by_job['cpa'].quantile(0.33)
    else C['amber'] if v <= cpa_by_job['cpa'].quantile(0.66)
    else C['no']
    for v in cpa_by_job['cpa']]
roi_colors = [
    C['yes'] if v >= cpa_by_job['roi_pct'].quantile(0.66)
    else C['amber'] if v >= 0
    else C['no']
    for v in cpa_by_job['roi_pct']]

fig.add_trace(go.Bar(y=cpa_by_job['job'],x=cpa_by_job['cpa'],
    orientation='h',
    marker_color=cpa_colors,
    text=cpa_by_job['cpa'].apply(lambda x: f'GBP{x:.0f}'),
    textposition='outside',
    textfont_size=9,showlegend=False,), row=1, col=1)

fig.add_trace(go.Bar(y=cpa_by_job['job'],x=cpa_by_job['roi_pct'],
    orientation='h',
    marker_color=roi_colors,
    text=cpa_by_job['roi_pct'].apply(lambda x: f'{x:.0f}%'),
    textposition='outside',textfont_size=9,
    showlegend=False,), row=1, col=2)

fig.add_vline(x=0, line_dash='solid', line_color=C['dark'],
    line_width=1, row=1, col=2)

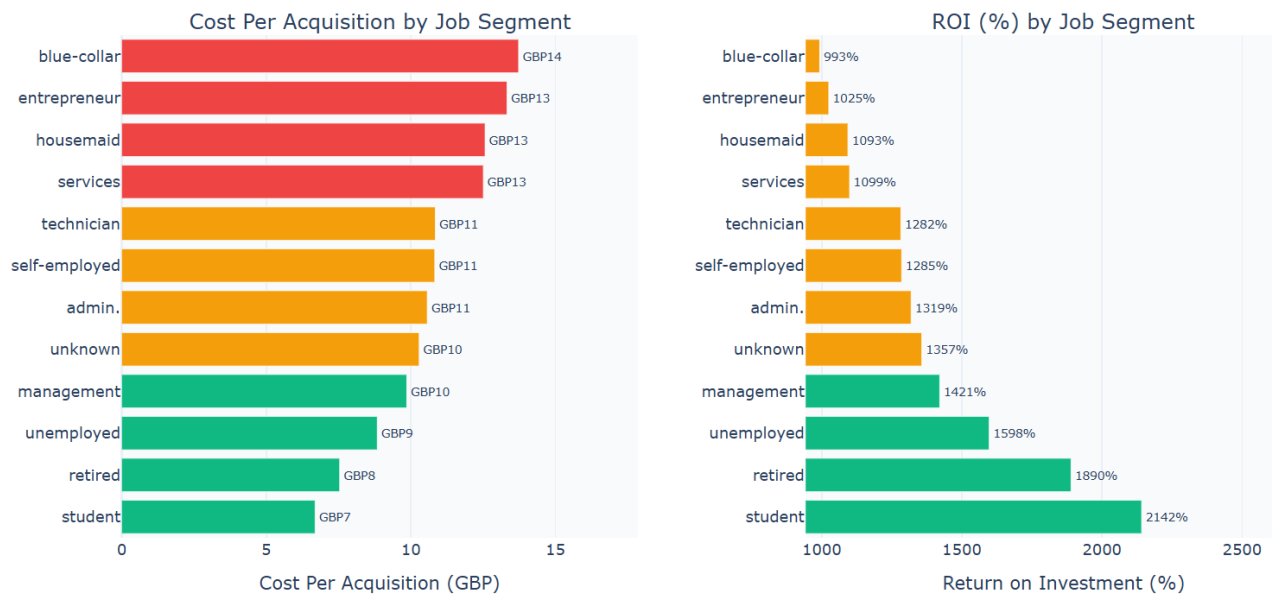
fig.update_layout(height=520,title_text='Campaign ROI Analysis by Customer Segment',
    title_font_size=15,paper_bgcolor='white',
    plot_bgcolor=C['bg'],uniformtext_minsize=8,
    uniformtext_mode='hide',margin=dict(t=80, b=40, l=20, r=80),)

# Extend x-axis range to give outside Labels room
cpa_max = cpa_by_job['cpa'].max()
roi_max = cpa_by_job['roi_pct'].max()
roi_min = cpa_by_job['roi_pct'].min()

fig.update_xaxes(title_text='Cost Per Acquisition (GBP)',
    range=[0, cpa_max * 1.3],
    row=1, col=1)
fig.update_xaxes(title_text='Return on Investment (%)',
    range=[min(roi_min * 1.3, roi_min - 50), roi_max * 1.3],
    row=1, col=2)
fig.show()

```


Campaign ROI Analysis by Customer Segment



```

duration_bins = [0, 60, 120, 180, 240, 300, 420, 600, 900, 9999]
duration_labels = ['<1m', '1-2m', '2-3m', '3-4m', '4-5m', '5-7m', '7-10m', '10-15m', '15m+']

df['duration_bucket'] = pd.cut(
    df['duration'],
    bins=duration_bins,
    labels=duration_labels,
    right=True
)

duration_analysis = (
    df.groupby('duration_bucket', observed=True)
    .agg(
        contacts = ('deposit_binary', 'count'),
        conversion_rate = ('deposit_binary', 'mean'),
    )
    .reset_index()
)

duration_analysis['conversion_rate'] = (duration_analysis['conversion_rate'] * 100).round(

fig, ax = plt.subplots(figsize=(14, 5))
fig.patch.set_facecolor('white')

bar_colors = [
    C['yes'] if v >= duration_analysis['conversion_rate'].quantile(0.66)
    else C['amber'] if v >= duration_analysis['conversion_rate'].quantile(0.33)
    else C['no']
    for v in duration_analysis['conversion_rate']
]

```

```

bars = ax.bar(
    duration_analysis['duration_bucket'].astype(str),
    duration_analysis['conversion_rate'],
    color=bar_colors, edgecolor='none', width=0.7
)

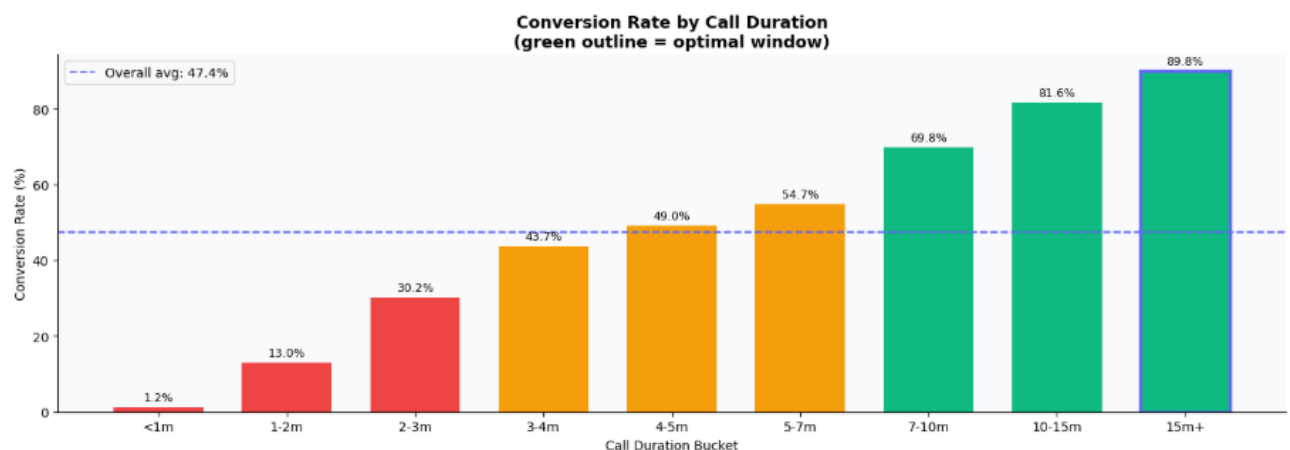
# Highlight optimal window
optimal_idx = duration_analysis['conversion_rate'].idxmax()
optimal_bucket = duration_analysis.loc[optimal_idx, 'duration_bucket']
ax.bar(
    str(optimal_bucket),
    duration_analysis.loc[optimal_idx, 'conversion_rate'],
    color=C['yes'], edgecolor=C['primary'], linewidth=2.5, width=0.7
)

ax.axhline(df['deposit_binary'].mean()*100, color=C['primary'],
           linestyle='--', linewidth=1.5,
           label=f'Overall avg: {df["deposit_binary"].mean()*100:.1f}%')
ax.set_title('Conversion Rate by Call Duration\n(green outline = optimal window)',
             fontsize=13, fontweight='bold')
ax.set_xlabel('Call Duration Bucket')
ax.set_ylabel('Conversion Rate (%)')
ax.bar_label(bars, fmt='%.1f%%', padding=3, fontsize=9)
ax.legend(fontsize=10)
ax.set_facecolor(C['bg'])

plt.tight_layout()
plt.savefig('/kaggle/working/output/optimal_window.png', dpi=120, bbox_inches='tight')
plt.show()

print(f'Optimal call duration window: {optimal_bucket}')
print(f'Conversion rate at optimum: {duration_analysis.loc[optimal_idx, "conversion_rate"]}')
print(f'vs overall average: {df["deposit_binary"].mean()*100:.1f}%')

```



Optimal call duration window: 15m+
Conversion rate at optimum: 89.8%
vs overall average: 47.4%

CHAPTER 5

CONCLUSION

Several important insights into consumer behavior and campaign performance were uncovered by the thorough examination of the Bank Marketing Dataset:

Effect of Communication Channels: The cellular contact approach performs noticeably better than traditional telephone lines, according to the simulated A/B test. The cellular group's conversion rate was significantly greater, indicating that mobile outreach is more successful at contacting clients when they are most likely to interact.

The "Duration" Factor: According to statistical study, the best indicator of a successful subscription is call duration. The length of the conversation and the probability of a "Yes" response are positively correlated, suggesting that interaction quality is a key factor in conversion.

Customer Financial Profile: Term deposit subscriptions were more common among customers with larger average annual balances. On the other hand, people who already had personal or home loans were less inclined to commit to new long-term investment products, probably because they had less money to spend.

Seasonal Patterns: The temporal study revealed that the effectiveness of marketing activities varies throughout the year. Subscription rates spiked in several months, particularly around the end of the fiscal quarter, indicating that timing is a key factor in consumer financial decision-making.

Demographic Hotspots: The data showed that, while having a lower volume, certain occupational categories, such as "retired" and "student" populations, frequently had greater conversion rates than high-volume "blue-collar" segments.

RECCOMANDATIONS

1. Prioritise Cellular Over Telephone

Cellular contact converts at a statistically significant higher rate, according to the A/B test. If at all possible, direct the outgoing campaign spending toward cellular. The various chemicals at scale.

2. Educate Agents on the Best Call Window

A particular call length range that maximizes conversion is found using the duration analysis. Too-short calls don't end. Long calls are wasted on uncooperative clients. As a useful coaching tool, share this discovery with the telesales staff.

3. Limit the number of contacts per campaign to five.

After five encounters, the research indicates diminishing returns, and after that, the conversion rate is actively harmed. To avoid wasting money and lower the possibility of consumer annoyance, set a hard cap of five contacts per customer every campaign.

4. Budget should be allocated by CPA rather than conversion rate.

In comparison to segments with moderate conversion rates, some have higher conversion rates but also higher contact expenses, which results in a lower real ROI. Instead of focusing on headline conversion rate, reframe budget allocation decisions around cost per acquisition.

5. Concentrate on March and September for Spring Campaigns

Seasonal patterns are seen in the monthly analysis. Campaigns that are started during these months routinely do better than others. Campaign calendars should be aligned appropriately.

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